T14

$\xi \sim N(a, \sigma_x^2)$ $\eta \sim N(b, \sigma_y^2)$ $\sigma_x^2 = 2$ $\sigma_y^2 = 1$ $\vec{x}_n = [-1{,}44; -0{,}1; 2{,}42]$

$\alpha = 0{,}05$ $n = 3$ $m = 2$ $\vec{y}_m = [-2{,}29; -2{,}91]$

$H_0: a = b$ $H_1: a > b$

$\frac{\overline{x} - a}{\sigma_x}\sqrt{n} \sim N(0;1)$ $\frac{\overline{y} - b}{\sigma_y}\sqrt{m} \sim N(0;1)$ (по т. Фишера)

$$\bar{x}-a \sim N\left(0; \frac{\sigma_x^2}{n}\right)$$

$$\bar{y}-b \sim N\left(0; \frac{\sigma_y^2}{m}\right) \to \bar{x}-\bar{y}-(a-b) \sim N\left(0; \frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}\right) \to \Delta = \frac{\bar{x}-\bar{y}-(a-b)}{\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}} \sim N(0;1)$$

$G: \Delta \geq u_{1-\alpha}$ $\qquad \Delta = \dfrac{\bar{x}-\bar{y}}{\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}} \sim N(0;1)$

Пусть $a-b=\theta \to \bar{x}-\bar{y} \underset{H_1}{\sim} N\left(\theta, \frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}\right) \to W = P(\tilde{\Delta} \in G | H_1) = P(\tilde{\Delta} \geq$

$\geq u_{1-\alpha}\left(\theta, \frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}\right))$